

Folding and Self-Assembly of the TatA Translocation Pore Based on a Charge Zipper Mechanism

SCIFACT-EVIDENCE-11419230

Evidence abstract

We propose a concept for the folding and self-assembly of the pore-forming TatA complex from the Twin-arginine translocase and of other membrane proteins based on electrostatic "charge zippers.

" Each subunit of TatA consists of a transmembrane segment, an amphiphilic helix (APH), and a C-terminal densely charged region (DCR).

The sequence of charges in the DCR is complementary to the charge pattern on the APH, suggesting that the protein can be "zipped up" by a ladder of seven salt bridges.

The length of the resulting hairpin matches the lipid bilayer thickness, hence a transmembrane pore could self-assemble via intra- and intermolecular salt bridges.

The steric feasibility was rationalized by molecular dynamics simulations, and experimental evidence was obtained by monitoring the monomer-oligomer equilibrium of specific charge mutants.

Similar "charge zippers" are proposed for other membrane-associated proteins, e.g., the biofilm-inducing peptide TisB, the human antimicrobial peptide dermcidin, and the pestiviral E(RNS) protein.

Document metadata

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Document ID | 11419230